$$P_{total} \times V_{total} = 1 \text{ atm} \times 22.4\ l$$

$$\therefore P_{total} = \frac{1 \text{ atm} \times 22.4\ \cancel{l}}{5\ \cancel{l}}$$

$$P_{total} = 4.48 \text{ atm}$$

$$\therefore P_{O_2} = 0.21 \times 4.48 \text{ atm}$$

$$= 0.94 \text{ atm}$$

$$P_{He} = x_{He} \times P_{total}$$

$$x_{He} = \frac{n_{He}}{n_{O_2} + n_{He}}$$

$$= \frac{2.05}{0.54 + 2.05}$$

$$x_{He} = \frac{2.05}{2.59} = 0.79$$

$$\therefore P_{He} = 0.79 \times 4.48 \text{ atm}$$

$$P_{He} = 3.54 \text{ atm}$$

6 (b) $2KClO_{3\,(s)} \longrightarrow 2KCl_{(s)} + 3O_{2\,(g)}$

$P_{total} = 772$ mm Hg

$P_{H_2O} = 26.7$ mm Hg

$$P_{total} = P_{O_2} + P_{H_2O}$$

$$\therefore P_{O_2} = P_{total} - P_{H_2O}$$

$P_1 = 26.7$ mm Hg $\quad T_2 = 295$ K

$T_1 = 300$ K $\quad P_2 = ?$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\Rightarrow P_2 = \left(\frac{P_1}{T_1}\right) T_2 = \frac{26.7 \text{ mm Hg}}{300\ \cancel{K}} \times 295\ \cancel{K}$$

$$P_2 = 26.26 \text{ mm Hg}$$

$$\therefore P_{O_2} = 772 - 26.26$$

$$= 745.74 \text{ mm Hg}$$

7) $t_1 = 1.5$ minutes $\quad (gas)_{\text{hydro carbon}}$

$t_2 = 4.73$ minutes $\quad (gas)_{\text{Bromine}}$

$$\frac{\gamma_{Hydrocarbon}}{\gamma_{Bromine}} = \frac{t_{Bromine}}{t_{Hydrocarbon}}$$

($\because$ Volume is constant)

$$= \frac{4.73 \text{ minutes}}{1.5 \text{ minutes}}$$

$$= 3.15$$

$$\frac{\gamma_{Hydrocarbon}}{\gamma_{Bromine}} = \sqrt{\frac{M_{Bromine}}{M_{hydrocarbon}}}$$

$$3.15 = \sqrt{\frac{159.8 \text{ gmol}^{-1}}{M_{hydro\ carbon}}}$$

Squaring on both sides and rearranging,

$$M_{hydro\ carbon} = \frac{159.8 \text{ g mol}^{-1}}{(3.15)^2}$$

$M_{hydro\ carbon} = 16.1 \text{ g mol}^{-1}$

$n(12) + (2n+2)\ 1 = 16$ $\quad$ [$\because$ general formula for hydrocarbon C_nH_{2n+2} (alkane)]

$12n + 2n + 2 = 16$

$14n = 16 - 2$

$14n = 14$

$n = 1$

$\therefore$ The hydro carbon is $C_1H_{2(1)+2} = CH_4$

8) Critical temperature of a gas is defined as the temperature above which it cannot be liquified even at high pressures.

$\therefore$ When cooling starts from 700 K, H_2O will liquify first, then followed by ammonia and finally carbondioxide will liquify.

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